

Benchmarking Complex Instruction-Following with Multiple Constraints Composition

Bosi Wen^{1,*} Pei Ke^{3,*} Xiaotao Gu² Lindong Wu² Hao Huang² Jinfeng Zhou¹
 Wenchuang Li^{4,†} Binxin Hu^{5,†} Wendy Gao² Jiaxin Xu¹ Yiming Liu¹
 Jie Tang¹ Hongning Wang¹ Minlie Huang^{1,‡}

¹Tsinghua University ²Zhipu AI ³University of Electronic Science and Technology of China

⁴China University of Geosciences ⁵Central China Normal University
 wbs23@mails.tsinghua.edu.cn, aihuang@tsinghua.edu.cn

Abstract

Instruction following is one of the fundamental capabilities of large language models (LLMs). As the ability of LLMs is constantly improving, they have been increasingly applied to deal with complex human instructions in real-world scenarios. Therefore, how to evaluate the ability of complex instruction-following of LLMs has become a critical research problem. Existing benchmarks mainly focus on modeling different types of constraints in human instructions while neglecting the composition of different constraints, which is an indispensable constituent in complex instructions. To this end, we propose COMPLEXBENCH, a benchmark for comprehensively evaluating the ability of LLMs to follow complex instructions composed of multiple constraints. We propose a hierarchical taxonomy for complex instructions, including 4 constraint types, 19 constraint dimensions, and 4 composition types, and manually collect a high-quality dataset accordingly. To make the evaluation reliable, we augment LLM-based evaluators with rules to effectively verify whether generated texts can satisfy each constraint and composition. Furthermore, we obtain the final evaluation score based on the dependency structure determined by different composition types. COMPLEXBENCH identifies significant deficiencies in existing LLMs when dealing with complex instructions with multiple constraints composition¹.

1 Introduction

Large language models (LLMs) have proven their remarkable abilities in addressing various NLP tasks [1]. Among these, instruction following is one of the most crucial requirements for LLM applications as it determines how well LLMs align with human intents [2]. In real-world use of LLMs, almost all the tasks are formulated as instruction following, where human instructions impose different constraints on the model output to specify the requirement of specific tasks [3].

Hence, how to accurately measure the quality of instruction following has become an essential problem. While early works focused on simple and direct human instructions in traditional NLP tasks, such as translation and text classification [4, 5, 6], recent works have resorted to complex instructions consisting of multiple constraints [3, 7, 8, 9], which are important constituents of LLM’s real-world use including role-play [10] and LLMs as agents [11]. These complex instruction-

*Equal contribution

†Work done when these authors interned at Zhipu AI.

‡Corresponding author

¹Our dataset and codes are available at <https://github.com/thu-coai/ComplexBench>.

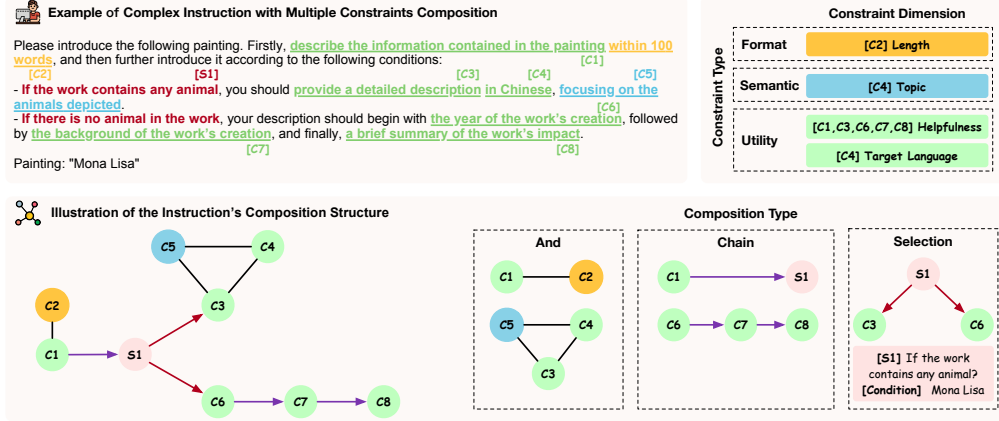


Figure 1: An example instruction of COMPLEXBENCH. All constraint dimensions contained in the instruction are marked with underlines and colors, which are categorized into three constraint types in our taxonomy: Format, Semantic, and Utility. Below is the composition structure of the instruction, where these constraint dimensions are combined through three composition types: *And*, *Chain*, and *Selection*.

following benchmarks aim to measure whether the generated text can meet every constraint in the input instruction.

However, we argue that existing complex instruction-following benchmarks neglect to model the composition of constraints, causing insufficient evaluation of the LLMs' ability to follow complex instructions. Since composition is a natural phenomenon in language use and a long-standing research problem in the NLP community [12, 13, 14, 15], it is a necessary ingredient in complex instructions to specify structural combinations of different constraints. In addition, the ignorance of composition leads to issues in both dataset construction and evaluation method design. On dataset construction, existing benchmarks are currently limited to simple composition types such as *And* which represents coordination between different constraints [3]. As shown in Figure 1, in addition to *And*, complex instructions can also include more intricate composition types of constraints, such as *Chain* (for sequential completion of constraints) and *Selection* (for conditional selection of constraints). Regarding evaluation method design, incorporating more complex composition types brings challenges in both constraint / composition evaluation and final score aggregation. First, complex instructions with structural combinations of constraints make it hard to evaluate each constraint / composition type independently with LLMs / rules due to their coupling. Then, simple aggregation methods for each constraint result, such as direct averaging, which is commonly adopted by existing benchmarks neglect the dependency among constraints brought by composition, causing potential biases in evaluation results.

In this paper, we propose COMPLEXBENCH, a novel benchmark to comprehensively evaluate the ability of LLMs to follow complex instructions. COMPLEXBENCH is manually constructed based on a hierarchical taxonomy of complex instructions, including 4 constraint types, 19 constraint dimensions, and 4 composition types, which provide a broad perspective to assess the performance of LLMs in dealing with complex instructions. To precisely measure whether LLMs' generated texts satisfy all these constraints and composition types, we design a yes / no question to verify each constraint and composition type respectively, inspired by the existing works on QA-based evaluation [16, 17, 7]. Then, we propose a new evaluation method for complex instruction-following called rule-augmented LLM-based evaluation. This method first extracts evaluation segments from generated responses for each yes / no question and then solves each question with LLMs or rules. Finally, the answers to each question are aggregated via the dependency structure among these questions, which is built based on the composition types. COMPLEXBENCH accompanied by our proposed evaluation method is expected to systematically reveal the deficiencies of existing LLMs on complex instructions and provide insights on the improvement of LLMs when dealing with various constraints and compositions. Our main contributions are as follows:

- We propose a comprehensive hierarchical taxonomy for complex instructions, including 4 constraint types, 19 constraint dimensions, and 4 composition types. We manually collect

Benchmark	Data Size	Constraint Taxonomy	Composition Type				Evaluation Method		
			<i>And</i>	<i>Chain</i>	<i>Selection</i>	<i>Nested</i>	LLM-based	Rule-based	Aggregation Function
WizardLM Testset [18]	218	-	✓	-	-	-	✓	-	-
CELLO [8]	523	4	✓	✓	-	-	-	✓	Average
FollowBench [3]	820	5	✓	-	-	-	✓	✓	Average
IFEval [19]	541	25	✓	-	-	-	-	✓	Average
InfoBench [7]	500	5	✓	-	-	-	✓	-	Average
CoI Testset [20]	1,068	-	-	✓	-	-	-	✓	-
COMPLEXBENCH (ours)	1,150	4-19	✓	✓	✓	✓	✓	✓	Dependency-based Aggregation

Table 1: Comparisons between COMPLEXBENCH and other benchmarks, illustrating the features including dataset sizes, constraint taxonomies, composition types, and evaluation methods. - in Aggregation Function means there is no step to evaluate each constraint and aggregate the final score.

a high-quality benchmark dataset for complex-instruction following, covering all types of constraints and compositions in our taxonomy.

- We accompany the benchmark with a new automated evaluation method to accurately evaluate the ability of LLMs to follow complex instructions, which integrates the advantages of LLM-based and rule-based methods to verify each constraint and composition type and aggregates the final score via the dependency structure brought by composition types.
- We conduct experiments on the proposed benchmark for a wide range of established LLMs, systematically revealing their deficiencies on various constraints and compositions.

2 Related Work

Evaluation of Instruction-Following. Instruction following remains one of the most important factors determining the practicality of LLMs [21]. Therefore, numerous studies have attempted to evaluate it from various aspects. Earlier works used to focus on simple human instructions formed with mostly a single constraint, such as semantic [5, 4, 6] and format [19, 22, 23] constraints. Since LLMs have been gradually applied to address complex real-world tasks, users have to form complex instructions, which naturally call for the evaluation of the LLMs’ ability in complex instruction following [3, 7]. WizardLM [18] employs two strategies, *In-Breadth Evolving* and *In-depth Evolving*, to form complex instructions from simple ones. CELLO [8] defines complex instructions from task descriptions and input text, and evaluates LLMs with real-world scenarios data. Unlike our work, which includes subjective and objective constraints and combines LLM-based and rule-based evaluations, CELLO focuses only on objective, rule-verifiable constraints and uses rule-based scoring functions for evaluation. Nonetheless, we argue that these benchmarks neglect to model the composition of constraints, which is an important character in complex instructions and brings non-negligible structural complexity that is crucial to assessing LLMs’ abilities.

Compositionality in NLP. Previous studies have explored compositionality across traditional NLP tasks, including semantic parsing [24, 25, 26], machine translation [26, 27], style transfer [28], and data-to-text generation [29]. However, in the task of instruction-following, how the LLMs deal with the compositionality in instructions is still under-explored. CompMCTG [30] investigates the compositionality of multiple control attributes for LLMs, which is a topic neighboring ours. Nevertheless, our work studies more complex composition types beyond simple coordination between different constraints, such as *Chain* and *Selection* and their nested structures, which form the basis of many real-world complex tasks for LLMs.

3 COMPLEXBENCH Framework

3.1 Overview

To comprehensively evaluate the ability of LLMs to follow complex instructions, we propose a hierarchical taxonomy to define constraints and composition types. For constraints, we extend common constraints in controlled text generation tasks to the instruction-following tasks and consider a two-level structure including coarse-grained types and fine-grained dimensions (Section 3.2). As for compositions that indicate structural combinations of constraints, we consider the characteristics of instruction-following tasks to define the composition types according to existing works on compositionality in traditional NLP tasks (Section 3.3).